OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 02

Team No.: 19

Project Title: Kernel Monitor

Team Members

Roll Number	Student Name	Signature
2520030184	Aniketh Pani Kakanur	
2520030558	Rohaan Jacob Variganti	
2520030053	V.Karthik	

1. Abstract

Kernel Monitor is a Linux-based system monitoring application developed to provide real-time information about system resources and running processes. The project uses the Linux /proc filesystem and POSIX/Linux system calls to collect information such as CPU utilization, memory and swap usage, system uptime, load average, kernel version, and process details. It also provides process management features using signals such as SIGTERM, SIGKILL, SIGSTOP, and SIGCONT. The application tracks process creation and termination events and displays them through an interactive terminal-based interface using the ncurses library. The project demonstrates important Operating Systems and Systems Programming concepts including process management, CPU and memory monitoring, signals, file descriptors, the /proc filesystem, and Linux system calls. The expected outcome is a lightweight and user-friendly Linux monitoring tool that helps users observe system activity and understand how operating system resources and processes can be accessed and managed from user space.

2. Problem Statement

Linux systems contain large amounts of information about system resources and running processes, but accessing and understanding this information directly through system interfaces can be difficult. The project aims to develop a simple terminal-based Kernel Monitor that collects and displays real-time system and process information using the Linux /proc filesystem, POSIX APIs, and system calls, while demonstrating important Operating Systems and Systems Programming concepts.

3. Objectives

1. To monitor CPU, memory, swap, uptime, load average, and other system resources in real time.

2. To display detailed information about running processes, including CPU usage, memory usage, state, threads, and process relationships.
3. To provide basic process control and event tracking using Linux signals and system calls.
4. To demonstrate Linux /proc filesystem, POSIX APIs, process management, signals, and ncurses through an interactive monitoring tool.

4. Proposed Methodology

The project will be implemented in C++ on Linux using the /proc filesystem, POSIX APIs, Linux system calls, and the ncurses library. System and process information will be collected and processed to display CPU, memory, process, and event details through an interactive terminal interface. Linux signals and system calls will be used for process control and monitoring.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
/proc Filesystem	Reads CPU, memory, uptime, and process information.
Process Management	Monitors PIDs, process states, threads, and relationships.
kill() / Signals	Controls processes using SIGTERM, SIGKILL, SIGSTOP, and SIGCONT.
open(), read(), close()	Reads system information from /proc.
opendir(), readdir(), closedir()	Scans process directories under /proc.
ncurses / POSIX APIs	Provides the terminal interface and input handling.

6. Individual Contribution

(Clearly specify the responsibility of each team member in the project.)

Roll Number	Student Name	Individual Responsibility
2520030184	Aniketh Pani Kakanur	System monitoring and /proc data collection.
2520030558	Rohaan Jacob Variganti	Process monitoring and process information.
2520030053	V.Karthik	Process control, UI, and testing.

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating environment for developing and running the Kernel Monitor.
C / C++	Programming language and compiler used to implement the project.
ncurses	Used to create the interactive terminal-based user interface.
POSIX / Linux APIs	Used for file operations, directory operations, signals, system information, and process control.
GNU Make / Makefile	Used to compile and build the project.

8. Expected Outcome

The project is expected to provide a lightweight Linux-based monitoring tool that displays real-time CPU, memory, system, and process information through an interactive terminal interface. It will allow users to view process details, monitor process events, and perform basic process control using Linux signals. The project will demonstrate practical Operating Systems and Systems Programming concepts using the /proc filesystem, POSIX APIs, Linux system calls, and ncurses.

9. GitHub Repository

GitHub Repository Link: https://github.com/Cyanexani/2520030184_053_558-OSSP

Repository Created:

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: Dr. K. Anusha

Signature: ____

Remarks: ____

Faculty Use

Project Approved: __________________________________

☐ Yes ☐ No

Date: ____

Faculty Signature: ____